INCH-POUND
MIL-PRF-49177B(CR)
19 MAY 1997
SUPERSEDING
MIL-S-49177A(CR)
8 April 1982

PERFORMANCE SPECIFICATION
SCAN-INTERLACE, PL-1408/UA

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers one type of common module, Scan-Interlace, PL-1408/UA (20 to 62 Hz).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-5855

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	- Standard General Requirements for Electronic Equipment
MIL-HDBK-338/1	- Electronic Design Reliability
MIL-HDBK-781	- Reliability Testing for Engineering Development, Qualification & Production

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings, and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2). Drawings listed below are to be used to ensure accurate interface and interconnection of the item produced under this performance specification with existing hardware.

DRAWINGS

USA CECOM

SM-C-773541	- Connector, Receptacle, Electrical
SM-C-773501	- Connector, Receptacle, Electrical
SM-D-773944	- Schematic, Scan and Interlace
SM-D-773945	- Printed Wiring Board, Scan and Interlace
SM-D-773946	- Scan-Interlace, PL-1408/UA

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

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ASTM E 122 - Sample Size to Estimate a Measure of Quality for a Lot or Process

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z.1.4 - Sampling Procedures; Tables for Inspection by Attributes

ANSI/IPC-S-815A - General Requirements for Soldering Electronic Interconnections

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, N.Y., N.Y. 10018).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

3.2 Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Scan-Interlace PL-1408/UA consists of a Printed Wiring Board (PWB) and circuitry to drive a mechanical scanner at scan rates from 20 to 62 Hz. It is capable of accepting an external trigger from another source to synchronize scanner operation with a vidicon camera.

3.3.1 Design. The Scan-Interlace shall meet the form, fit, function and interface requirements as specified herein.

3.3.2 Weight. The weight of the Scan-Interlace shall not be greater than 0.35 (+0.04 -0.08) pound as verified by a balance or scale with not less than ±1% accuracy.

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3.3.3 Dimensions. The Scan-Interlace PWB shall be of appropriate material with a nominal thickness of 0.062 inches and shall have dimensions of 3.000 inches x 6.000 inches, ± 0.010 inches. When mounted, components shall not extend more than a maximum of 0.55 inches from the surface of the printed wiring board. All four corners of the board shall be trimmed at a 45° angle between two points 0.12 inches vertical and horizontal of the corners, removing a triangle of board material 0.12 by 0.12 on either side of the right angle (corner). The x-axis datum line shall be located parallel to the lower edge of the board and 0.150 inches above that edge. It shall pass through the two 0.062 inch diameter holes located 0.150 inches above the edge and 0.200 in from the trimmed corners. The y-axis datum line shall be located 0.200 inches in from the left-hand edge of the board. It shall pass through the two left-hand 0.062 inch diameter holes located 0.200 inches in from the left edge and shall be perpendicular to the x-axis datum line. Two mounting holes 0.140 inches in diameter shall be located 2.625 inches above the x-axis datum line and 0.225 inches and 5.375 inches to the right of the y-axis datum line as shown on Drawing SM-D-773945. Three connectors shall be attached to the component side of the board for the purpose of interfacing with existing hardware. One connector shall have the dimensions of Part Number SM-C-773541-5 as shown on Drawing SM-C-773541, while the other two connectors shall have the dimensions of Part Number SM-C-773501-2 as shown on Drawing SM-C-773501. These connectors shall be located approximately as shown on Drawing SM-D-773946 for P1, J1 and J2. Six Tip Jacks shall also be attached to the component side of the board. These tip jacks shall mount in the pairs of 0.055 inch diameter holes located near the upper edge of the board. The Tip Jacks shall be JOLO part numbers LPCJ-419-05 through LPCJ-419-10 or M39024/11-05 through M39024/11-10 or the equivalent. They shall be located as shown on Drawing SM-D-773946. The relationship between their reference designations and their find numbers and colors is shown on SM-D-773946. The pinout assignments for the connectors shall be in accordance with Drawing SM-D-773944 to ensure proper interface with the existing hardware.

3.4 Components. The Scan-Interlace shall contain manufactured components selected by the contractor as required to meet the performance specifications contained herein.

3.5 Performance characteristics. Scan-Interlace characteristics specified herein apply to 42 Hz or 60 Hz operation. Unless otherwise specified (see 6.2), only 42 Hz operation applies. The voltages and waveforms required to measure performance characteristics are shown in Figures 3 through 8. Measurements shall be taken after these voltage and waveform preparations have been completed.

3.5.1 Power consumption. The Scan-Interlace shall operate with the following voltages and currents (see Figure 3).

a. Plus 5.0, ±0.3 Vdc (1.0 ampere peak) 0.65 rms watts maximum

b. Minus 5.0, ±0.3 Vdc (1.0 ampere peak) 0.25 rms watts maximum

c. Plus 15.0, ±0.5 Vdc and -15.0 ±0.5 Vdc (1.5 amperes peak) 6.6 rms watts maximum

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*Power shall be measured as the sum of the products of the power supply voltage and their average currents.

3.5.2 Identification resistor verification. The Scan-Interlace should contain two identification resistors as shown in block C-8 on Drawing SM-D-773944. Verify performance in accordance with 4.5.2

3.5.3 Regulator voltage. The Scan-Interlace should contain a positive voltage regulator of +2.40, ±0.15 Vdc and a negative voltage regulator of -2.40, ±0.15 Vdc. Verify compliance in accordance with 4.5.3.

3.5.4 Start-up protection. The Scan-Interlace shall be capable of electronically inhibiting interlace for 0.20, (+0.20-0.05) seconds in order to reduce current surge during start-up. As a check, turn off power with switch (1) as shown in Figure 3. Verify compliance in accordance with 4.5.4.

3.5.5 Scan transducer amplifiers. The Scan Position Transducer Amplifier and its adjacent stage shall produce a gain of 199.5, ±28.5 volts/volt for signals generated by the Scan Position Transducer. Verify compliance in accordance with 4.5.5.

3.5.6 Sync drive. The Scan-Interlace shall allow for adjusting the sync delay and the sync drive width electronics to obtain a 400, ±40 microsecond delay. This delay shall be between the sync drive waveforms and a 50% duty cycle (on the sync drive waveform) generated from the external input. Verify compliance in accordance with 4.5.6.

3.5.7 Spring limit pulse width. The Scan-Interlace shall provide a spring limit level adjustment that will allow setup of a spring limit pulse width of 4.0, ±0.6 milliseconds at 42 Hz or 2.50, ±0.25 milliseconds at 60 Hz. Verify compliance in accordance with 4.5.7.

3.5.8 Scan rate. The Scan-Interlace shall provide a scan rate waveform that is adjustable to an amplitude of 1.7, ±0.2 volts peak-to-peak (P-P) at 42 Hz or 2.0, ±0.2 volts (P-P) at 60 Hz. Verify compliance in accordance with 4.5.8.

3.5.9 Scan drive. The Scan-Interlace shall allow adjustment of the scan rate and the scan drive balance electronics to obtain a scan drive waveform with flat active times at an amplitude of 1.0, ±0.2 volt P-P. Verify compliance in accordance with 4.5.9.

3.5.10 Scan motor drive amplifier. The Scan-Interlace shall provide a gain between the scan drive waveform and the scan motors of 5.0, ±1.5 volts/volt. Verify compliance in accordance with 4.5.10.

3.5.11 Interlace transducer amplifier. The Scan-Interlace shall provide a gain of 100.0, ±4.0 volts/volt for signals generated by the Interlace Position Transducer. Verify compliance in accordance with 4.5.11.

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3.5.12 Interlace position reference. The Scan-Interlace shall provide the interlace position reference electronics capable of generating a waveform adjustable to an amplitude of 2.0, ±0.2 volts P-P. Verify compliance in accordance with 4.5.12.

3.5.13 Interlace rate reference. The Scan-Interlace shall provide an interlace position error gain adjustment that will provide an interlace position error waveform with an amplitude of 1.32, ±0.15 volts P-P at 42 Hz or 1.00, ±0.10 volt P-P at 60 Hz. Verify compliance in accordance with 4.5.13.

3.5.14 Interlace position error. The Scan-Interlace shall provide an interlace position error gain adjustment that will provide an interlace position error waveform with an amplitude of 1.32, ±.15 volts P-P at 42 Hz or 1.00, ±0.10 volt P-P at 60 Hz. Verify compliance in accordance with 4.5.14.

3.5.15 Solenoid drive amplifiers. The Scan-Interlace shall provide a gain between the interlace position error waveform and each solenoid drive amplifier output of 0.22, ±0.07 volts/volt. Verify compliance in accordance with 4.5.15.

3.5.16 Video gate. The Scan-Interlace shall provide video gate electronics which are capable of generating a video gate pulse width of 4.0, ±0.8 milliseconds, with an amplitude not less than 2 volts P-P, and in phase with the spring limits pulse at 42 Hz. The requirements at 60 Hz shall be the same except that the pulse width shall be 2.5, ±0.4 milliseconds. Verify compliance in accordance with 4.5.16.

3.5.17 Transducer bridge voltage. The Scan-Interlace shall provide a transducer bridge voltage which is adjustable to 1.9, ±0.1 Vdc. Verify compliance in accordance with 4.5.17.

3.5.18 Scan fail electronics. The Scan-Interlace shall contain the electronics to inhibit scan and interlace by applying an external voltage of + 5.0, ±0.1 Vdc to the scan disable input or by removing the scan position signal. Verify compliance in accordance with 4.5.18.

3.6 Environmental conditions.

3.6.1 Temperature shock. The Scan-Interlace shall not be damaged (see 6.5.1) by sudden changes in temperature between -54°C and +95°C.

3.6.2 High temperature. The Scan-Interlace shall not be damaged by storage to +95°C for 6 hours or continuos operation to +71°C, and shall operate for 30 minutes at +95°C. The scan motor HI waveform (P1-13, Figure 8) shall not change more than ±15 percent from the initial setup at ambient temperature (see 6.5.2) over the specified operating temperature range (+71°C). The solenoid drive amplifier output at J1-5 and J1-7 as shown in blocks B-5 and C-5 on Drawing SM-D-773944 shall not change more than ±15 percent from the initial setup at ambient temperature (see 4.4) over the specified operating temperature range (+71°C).

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3.6.3 Low temperature. The Scan-Interlace shall not be damaged by operation to -54°C, and storage to -62°C. The scan motor HI waveform (P1-13, Figure 8) shall not change more than ±15 percent from the initial setup at ambient temperature over the specified operating temperature range (-54°C). The solenoid drive amplifier output at J1-5 and J1-7 as shown in blocks B-5 and C-5 on Drawing SM-D-773944 shall not change more than ±15 percent from the initial setup at ambient temperature (see 4.4) over the specified operating temperature range (-54°C).

3.6.4 Shock. The Scan-Interlace shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.6.5 Vibration. The Scan-Interlace shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in Figure 1.

3.7 Reliability. The Scan-Interlace shall have a mean time between failure (MTBF) of least 17,500 hours, θ_0. Methods described in MIL-HDBK-338/1 and evaluated per MIL-HDBK-781 may be used. (see 4.7).

3.8 Coating. Unless otherwise specified, the Scan-Interlace and its components and parts shall be cleaned and coated to prevent corrosion, oxidation and fungus growth.

3.9 Preconditioning. Each Scan-Interlace shall be subjected to at least a 20-cycle preconditioning environment as specified in Figure 2 prior to Group A inspection (for exception to this requirement see 6.2).

3.10 Soldering. Unless otherwise specified, electronic soldering shall conform to ANSI/IPC-S-815A.

3.11 Nameplates and product marking. The assembly, parts, components, subassemblies and assemblies thereof shall be marked for identification and reference designation markings. For guidance, use MIL-HDBK-454.

3.12 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires or integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

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b. Conformance inspection (see 4.4)

4.1.1 Inspection parameters. Unless otherwise specified, the Scan-Interlace (PL-1408/UA) shall be tested at 42 Hz (see 6.2).

4.2 First article inspection. When specified in the contract, the first selected number of first article inspections shall be performed by the contractor on a sample unit (see 3.1). First article reliability testing per 4.7 will not be required unless specified in the contract or purchase order.

4.2.1 Inspections. All materials, parts, processes and assemblies shall be visually examined for conformance to the applicable specification or drawing. Inspections shall be made using ANSI/ASQ Z 1.4 sampling procedures. ASTM E 122 sample sizes shall conform to 89 as cited in the contract or purchase order.

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TABLE I. Visual inspections.

Inspection requirement	Requirement paragraph	Test paragraph
Any part or component missing or damaged	3.3, 3.4	3.3, 3.4
Weight not as specified	3.3.2	3.3.2
Coating not as specified	3.8	3.8
Soldering not as specified	3.10	3.10
Marking not as specified	3.11	3.11
Workmanship not as specified	3.12	3.12
Dimensions not as specified	3.3, 3.4	3.3, 3.4
Length		
Width		
Height (maximum)		
Solder/lead protrusion		
pins (connector location)		
Pins (guide location)		

4.2.2 Tests. Upon successful completion of inspections specified in 4.2.1, perform the tests of Table II in any order. Failure of any test shall constitute failure of first article inspection.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Power consumption	3.5.1	4.5.1
Identification resistor verification	3.5.2	4.5.2
Regulator voltage	3.5.3	4.5.3
Start-up electronics	3.5.4	4.5.4
Scan transducer amplifiers	3.5.5	4.5.5
Sync drive	3.5.6	4.5.6
Spring limit pulse width	3.5.7	4.5.7
Scan rate	3.5.8	4.5.8
Scan drive	3.5.9	4.5.9
Scan motor drive amplifier	3.5.10	4.5.10
Interlace transducer amplifier	3.5.11	4.5.11
Interlace position reference	3.5.12	4.5.12
Interlace rate reference	3.5.13	4.5.13
Interlace position error	3.5.14	4.5.14
Solenoid drive amplifiers	3.5.15	4.5.15
Video gate	3.5.16	4.5.16
Transducer bridge voltage	3.5.17	4.5.17
Scan fail electronics	3.5.18	4.5.18
Temperature shock	3.6.1	4.6.1
High temperature	3.6.2	4.6.2
Low temperature	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Reliability	3.7	4.7

4.2.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.g).

4.3 Inspection conditions. Unless otherwise specified herein, the Scan-Interlace shall meet the specified performance requirements at an ambient temperature of +23°C, ±5°C.

4.3.1 Input power. The Scan-Interlace shall operate with the following voltages and currents:

a. Plus 5.0, ± 0.3 volts direct current (Vdc) (1.0 ampere peak), 0.65 rms watt maximum

b. Minus 5.0, ±0.3 Vdc (1.0 ampere peak), 0.25 rms watt maximum

c. Plus 15.0, ±0.5 Vdc and -15.0, ±0.5 Vdc (1.5 amperes peak), 6.6 rms watts maximum

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NOTE: For 20 to 30Hz operation, the ±15 Vdc power supplies shall be reduced to ±5 Vdc and the total power required shall not be greater than 4.0 rms watts.

4.3.2 Supply ripple. The supply ripple shall not be greater than 0.1 volt P-P for the ±5 Vdc power supplies, and shall not be greater than 0.2 volt P-P for the ±5 Vdc power supplies, and shall not be greater than 0.2 volt P-P for the ±15 Vdc power supplies.

4.3.3 Supply rise time. The supply rise time shall not be greater than 0.5 seconds.

4.3.4 Supply tracking. Turn-on rise time of the ±15 Vdc power supplies shall not be greater than that of the ±5 Vdc power supplies. The rise times of the ±5 Vdc power supplies shall track within 20 percent.

4.3.5 Impedance. The supply output impedances shall not be greater than 0.1 ohm to 10 kHz and 5 ohms to 100 kHz.

4.4 Conformance inspection.

4.4.1 Visual Inspection. Inspection shall be the same as specified in 4.2.1.

4.4.2 Tests.

4.4.2.1 Group A inspection. Group A inspection shall be conducted at an ambient temperature of +23°C, ±5°C on all Scan-Interlaces following the preconditioning of 4.8 and the inspection of 4.4.1. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	On Scan-Interlace	
			Adjustment during test	Pins tested
Power consumption	3.5.1	4.5.1	None	None
Identification resistor verification	3.5.2	4.5.2	None	J1-1, J1-2, J1-3
±2.4 Vdc Regulator voltage	3.5.3	4.5.3	None	J1-18, J2-2
Start-up electronics	3.5.4	4.5.4	None	J1-17
Scan transducer amplifiers	3.5.5	4.5.5	None	J2-7, P1-15
Sync drive	3.5.6	4.5.6	Sync delay R106, Sync width R115	P1-30, P1-22
Spring limit pulse width	3.5.7	4.5.7	Spring limit level R18	J2-15
Scan rate	3.5.8	4.5.8	Scan rate R8	J2-13
Scan drive	3.5.9	4.5.9	Scan drive bal. R32 Scan rate R8	J2-6
Scan motor drive amplifier	3.5.10	4.5.10	None	J2-6, P1-13
Interlace transducer amplifier	3.5.11	4.5.11	None	P1-3, J1-9
Interlace position reference	3.5.12	4.5.12	Interlace pos. ref. R93	J1-8
Interlace rate reference	3.5.13	4.5.13	Interlace rate ref. R114	J1-13
Interlace position error	3.5.14	4.5.14	Pos. error gain R111	J1-14
Solenoid drive amplifiers	3.5.15	4.5.15	Solenoid drive balance R124	J1-14, J1-5, J1-7
Video gate	3.5.16	4.5.16	Gate width adjust R160	P1-21 J2-15
Transducer bridge voltage	3.5.17	4.5.17	Bridge voltage R122	P1-12
Scan fail electronics	3.5.18	4.5.18	None	P1-13, J1-5

4.4.2.2 Group B inspection. Not required.

4.4.2.3 Group C inspection. Group C inspections shall be conducted on Scan-interlace assemblies selected from units which have passed the tests in 4.4.2.1. The Sample (s) shall be

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subjected to the inspection listed in Table IV. The sample shall be selected in accordance with 4.4.2.3.1.

4.4.2.3.1 Sampling for Group C inspection. Group C inspection samples shall be selected as specified in the contract or purchase order.

TABLE IV Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Vibration	3.6.5	4.6.5

4.4.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract (see 6.2).

4.4.2.3.3 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the test of Table III.

4.4.2.4 Group D inspection. Not required.

4.5. Methods of inspection.

4.5.1 Power consumption. The assembly shall be set up according to 3.5.1 through 3.5.17. The average current being drawn by the assembly from each voltage source shall be measured. The power delivered shall be calculated as the sum of the products of the power supply voltage and their average currents, as found above. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Identification resistor verification. Measure the resistance between J1-1 to J1-2 for identification resistor No. 1 and between J1-3 to J1-2 for identification resistor No. 2 as shown on Drawing SM-D-773944. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Regulator voltage. Measure the +2.4 Vdc regulator voltage on J1-18 and the -2.4 Vdc regulator on J2-2 as shown on Drawing SM-D-773944. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Start-up electronics. Remove power from assembly by using switch (S1) as shown in Figure 3. Monitor J1-17 as shown on Drawing SM-D-773944 with an oscilloscope and measure the width of the pulse as power is re-applied to the assembly. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

4.5.5 Scan transducer amplifiers. Measure the scan position waveform amplitude at the input of the assembly (figure 4) using an oscilloscope. Also measure the scan position waveform amplitude after the second stage (J2-7 as shown on Drawing SM-D-773944). The following equation describes the gain of the scan transducer amplifier:

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Gain = voltage at $J2\text{-}7_{(P\text{-}P)}$/input voltage $_{(P\text{-}P)}$

Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Sync drive. Monitor the sync drive waveforms at P1-30 and P1-22 as shown on Sheet 1 of Drawing SM-D-773944 using an oscilloscope. P1-22 is the sync drive waveform generated from the external sync input. P1-30 is the sync drive waveform generated from the scan position waveform. The delay shall be measured from the positive going edge of P1-22 to the negative going edge of P1-30 after a 50 percent duty cycle has been achieved on P1-22. Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Spring limit pulse width. Monitor the spring limit pulse at J2-15 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the time between the negative going edge and the adjacent positive going edge of the pulse displayed on the oscilloscope. Failure to meet the requirements of 3.5.7 shall constitute failure of this test.

4.5.8 Scan rate. Monitor the scan rate waveform at J2-13 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the P-P amplitude of the scan rate waveform presented on the display. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.5.9 Scan drive. Monitor the scan drive waveform at J2-6 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the P-P voltage during the flat active time of the waveform displayed on the oscilloscope. Failure to meet the requirements of 3.5.9 shall constitute failure of this test.

4.5.10 Scan motor drive amplifier. Monitor the scan drive waveform of J2-6 and the scan motor output at P1-13 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the amplitude of the scan motor output waveform P-P and the scan motor output waveform P-P at a corresponding point during the flat active portion of the scan drive waveform. Measurements taken from the oscilloscope display shall be placed in the following gain equation:

Gain = Scan Motor Output / Scan Drive Output

Failure to meet the requirements of 3.5.10 shall constitute failure of this test.

4.5.11 Interlace transducer amplifier. Monitor the interlace position waveform at the input of the assembly (P1-3 and P1-2) and the output of the interlace position amplifier at J1-9 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the P-P amplitude of each waveform as presented on the oscilloscope display. The following equation defines the gain:

$$\text{Gain} = \frac{\text{interlace position output (J1-9)}_{(P\text{-}P)}}{\text{interlace position input (P1-3)}_{(P\text{-}P)}}$$

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Failure to meet the requirements of 3.5.11 shall constitute failure of this test.

4.5.12 Interlace position reference. Monitor the interlace position reference waveform at J1-8 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the P-P amplitude of the waveform as presented on the display. Failure to meet the requirements of 3.5.12 shall constitute failure of this test.

4.5.13 Interlace rate reference. Monitor the interlace rate reference waveform at J1-13 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the P-P amplitude of the waveform as presented on the oscilloscope display. Failure to meet the requirements of 3.5.13 shall constitute failure of this test.

4.5.14 Interlace position error. Monitor the interlace position error waveform at J1-14 as shown on Drawing 773944 using an oscilloscope. Measure the P-P amplitude during the flat action portion of the waveform as presented on the oscilloscope display. Failure to meet the requirements of 3.5.14 shall constitute failure of this test.

4.5.15 Solenoid drive amplifiers. Monitor the interlace position error waveform at J1-14 and both solenoid amplifier output waveforms at J1-5 and J1-7 (per SM-D-773944) using an oscilloscope. Adjust solenoid drive balance until output waveforms of both solenoid drive amplifiers have the same absolute amplitude. Measure the active portion of the interlace position error waveform and a corresponding point on both the solenoid No. 1 amplifier and the solenoid No. 2 amplifier. The gain of each amplifier can be described by the following equation:

$$\text{Gain} = \frac{\text{solenoid drive amplifier out (peak)}}{\text{interlace position error (peak)}}$$

Failure to meet the requirements of 3.5.15 shall constitute failure of this test.

4.5.16 Video gate. Monitor the video gate waveform at P1-21 and the spring limits pulse at J2-15 as shown on Drawing SM-D-773944 using an oscilloscope. Measure the P-P amplitude of the video gate waveform and its phase relationship to the spring limits pulse as presented on the oscilloscope display. Failure to meet the requirements of 3.5.16 shall constitute failure of this test.

4.5.17 Transducer bridge voltage. Monitor the transducer bridge voltage at P1-12 as shown on Drawing SM-D-773944. An average voltage shall be measured at this test point. Failure to meet the requirements of 3.5.17 shall constitute failure of this test.

4.5.18 Scan fail electronics. Monitor the scan motor drive output at P1-13 and the solenoid drive amplifier at J1-5 as shown on Drawing SM-D-773944 using an oscilloscope. Apply a +5.0, ±0.1 Vdc input (S3 ON) to the scan disable electronics (figure 8) and observe that signals on the oscilloscope display go to zero. Removal of signals from the display as voltage is

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applied to the scan disable input determines proper operation of the scan fail electronics. Remove the +5 Vdc input from the disable input and observe the return of signals on the display. While continuing to monitor the display, change the amplitude of the scan position signal at the input of the assembly (figure 5) (P1-15 and P1-14) to one-fourth of its original amplitude. The signal amplitude shall go to zero. Failure to meet the requirements of 3.5.18 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing shall be performed in accordance with the requirements of this specification, except when a contractor's plan showing equivalent methods has been approved by the contracting office. When the contractor's test plan deviates from the test procedures prescribed, the contractor shall be required to show where his own tests and methods for verifying the specified performance are substantiated. Also, measurements taken by the contractor during verification testing of all environmental requirements shall be accomplished using known commercially accepted testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device or similar equipment for which the technical capabilities to properly simulate extreme climatic or environmental conditions are scientifically documented. In general, the methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent or better than the recommended procedures found in 6.3.

4.6.1 Temperature shock. Measure compliance with the requirement of 3.6.1. It is recommended that the tests prescribed in 6.3 be performed. Failure to meet the requirement shall constitute failure of the sample.

4.6.2 High temperature (storage and operation). Measure compliance with the requirement of 3.6.2. It is recommended that the tests prescribed in 6.3 be performed. Failure to meet the requirement shall constitute failure of the sample.

4.6.3 Low temperature. Measure compliance with the requirement 3.6.3. It is recommended that the tests prescribed in 6.3 be performed. Failure to meet the requirement shall constitute failure of the sample.

4.6.4 Shock. Measure compliance with the requirement of 3.6.4. It is recommended that the tests prescribed in 6.3 be performed. Failure to meet the requirement shall constitute failure of the sample.

4.6.5 Vibration. Measure compliance with the requirement of 3.6.5. It is recommended that the tests prescribed in 6.3 be performed. Failure to meet the requirement shall constitute failure of the sample.

4.7 Reliability.

4.7.1 First article reliability. The lower MTBF (θ_1) shall be demonstrated in accordance with the following subparagraphs:

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4.7.1.1 Test level.

a. Temperature: +50°C, (+5 -0)°C

b. Temperature cycling: None

c. Vibration: -2.2 g, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be normal to the plan of the printed circuit board.

d. Equipment on-off cycling: Turn-ON and let temperature stabilize, hold for 3 hours, then turn OFF and let temperature stabilize. This cycle shall continue throughout the test.

4.7.1.2 Parameters. The discrimination ratio θ_0/θ_1shall be 3.

4.7.1.3 Failure criteria and test length. The test of Table III shall be performed at least once prior to and once after the test of 4.7.1.1. Additional testing of Table III may be selected by the contractor. The results of all Table III tests and any physical damage shall be used to determine the success or failure of the reliability test. In the event of a table III test failure, the time interval between the preceding successful test and the failure shall not contribute to the total cumulative hours. Testing shall continue until the total unit hours, together with the total count of relevant equipment failures, permit either an accept or reject decision in accordance with Figure 9. Only equipment “ON” time may contribute to the total unit hours. No single equipment’s “ON” time shall be less than one-half the average operating time of all equipment being tested.

4.8 Preconditioning. Each Scan-Interlace, while in a non-operating mode, shall be subjected to 20 cycles as specified in Figure 2. Preconditioning may be performed before or after conformal coating.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s Packaging Activity within the Military Department of Defense Agency, or within the Military Department’s system Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The Scan-Interlace, PL-1408/UA is intended for use in infrared systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required. Also include:

1) Time frame for submission of first article test reports when the first article tests are required.

2) Time frame for approval of first article test reports when first article tests are required.

3) First article reliability testing, if required.

c. Production delivery schedule.

1) Defined in terms of monthly lots.

2) Actions required relative to Group C failures (see 4.4.2.3.2).

d. Preservation and packaging (see section 5).

e. Methods used for verifying environmental performance criteria.

f. The preconditioning requirement 3.9 and inspection are required for procurement of all modules to this specification made directly for the Government. Implementation of 3.9 may be waived by systems contractors ordering modules to this specification for use in the manufacture of complete systems if these systems or subassemblies thereof are subjected to equivalent preconditioning.

g. Deliver all first article samples to the contracting officer.

h. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

i. Group C inspection monthly sample size to be established based on cumulative totals of concurrent contracts.

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j. Issue of the DoDISS to be cited in the solicitation, and if required, the specified issue of individual documents referenced (see 2.2.1 and 2.2.2).

6.3 Environmental inspection procedures. The test methods and procedures of MIL-STD-810 listed below are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the material to be tested.

Inspection	Spec. Paragraph	MIL-STD-810 Method	MIL-STD-810 Procedure	Remarks
Temperature shock	3.6.1	503.1	I	Steps 1 & 4 shall be +95°C and Step 2 shall be -54°C transition from +95°C to-54°C and from -54°C to +95°C shall include a 2 hour stabilizing period at +23°C.
High temperature (storage)	3.6.2	501.1		Storage temperature (Step 2) shall be +95°C for 6 hours duration.
High temperature (operation)	3.6.2	501.1	I	Continuos operation temperature (Step 4) shall be +71°C.
Low temperature (storage)	3.6.3	502	I	Storage temperature of Step 2 shall be -62°C for 9 hours.
Low Temperature (operation)	3.6.3	502	I	Operation temperature is -54°C.
Shock	3.6.4	516.2	II	Time duration: 11 milliseconds.
Vibration	3.6.5	514.2	VIII	Use curve V. As an alternative, use method 201A of MIL-STD-202.

6.5 Definitions.

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6.5.1 Damage. Breakage, loosening, shifting, evidence of corrosion or failure of any finish, hardware, connection or component; and any degradation of assembly performance to values less than specified.

6.5.2 Ambient temperature. Ambient temperature is 23°C ±5°C.

6.6 Subject term (key word) listing.

Infrared
Scan fail

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5855-0122)

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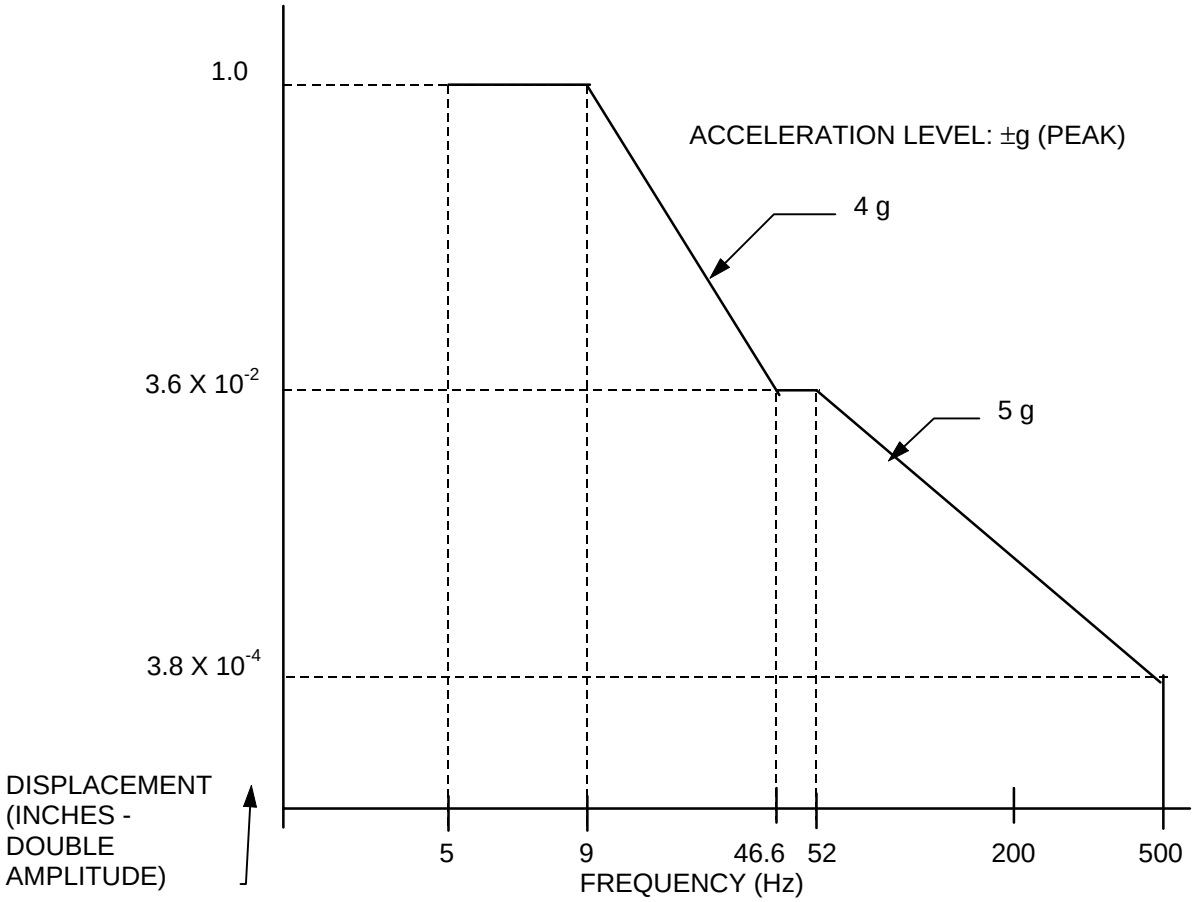


FIGURE 1. Vibration Test Profile

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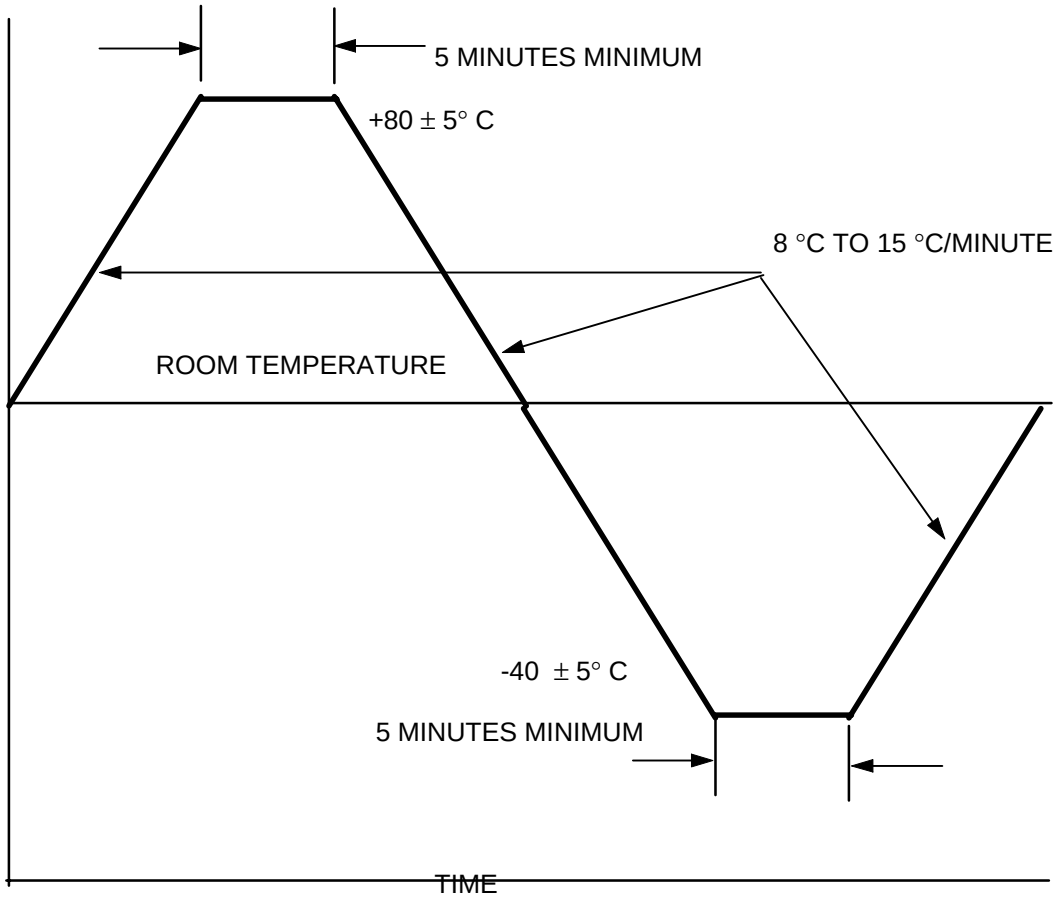


FIGURE 2. One Preconditioning Cycle

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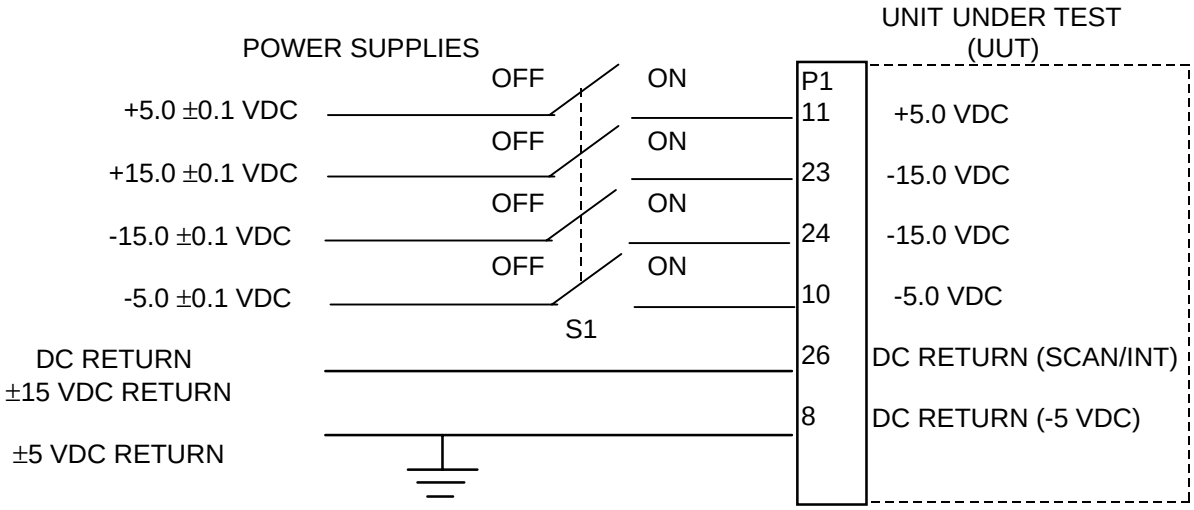


FIGURE 3. Test Interface.

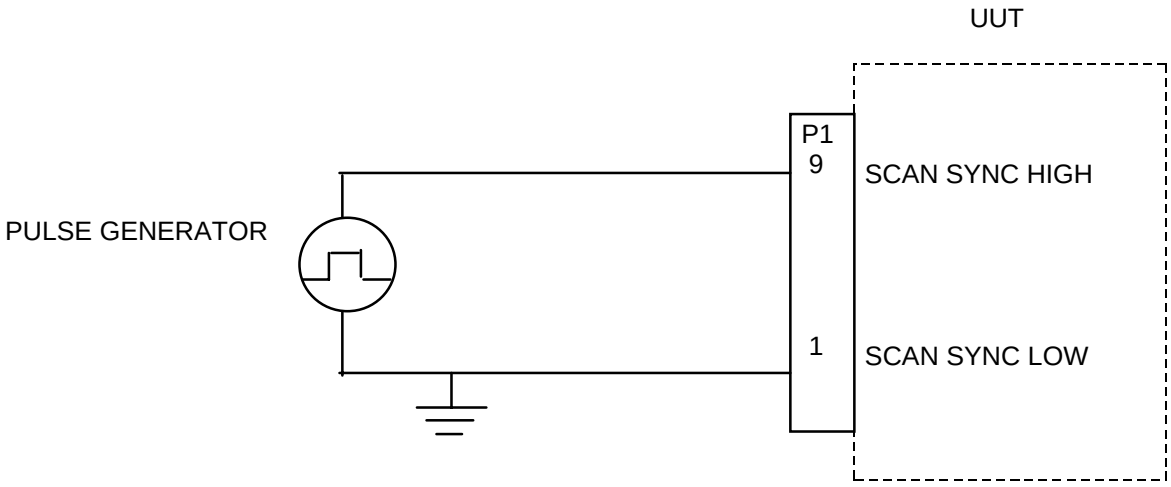


FIGURE 4. Test Interface.

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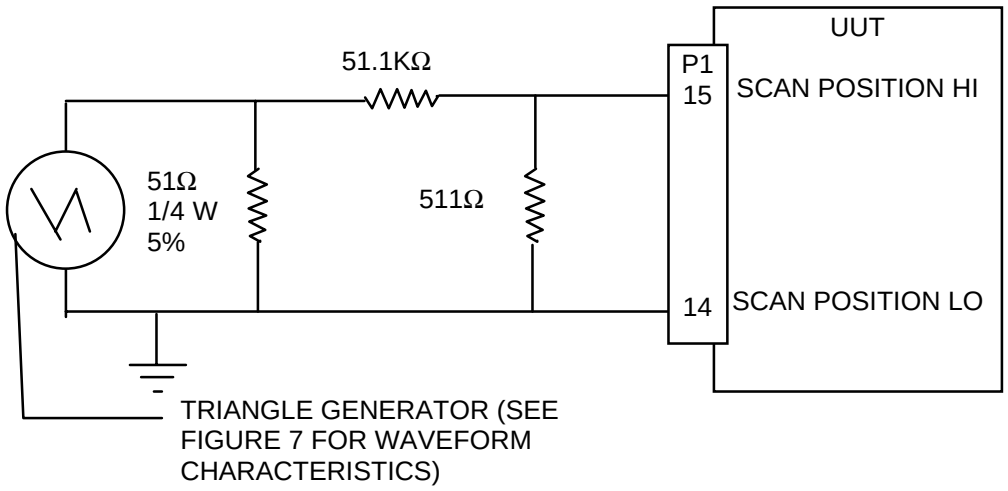


FIGURE 5. Test interface.

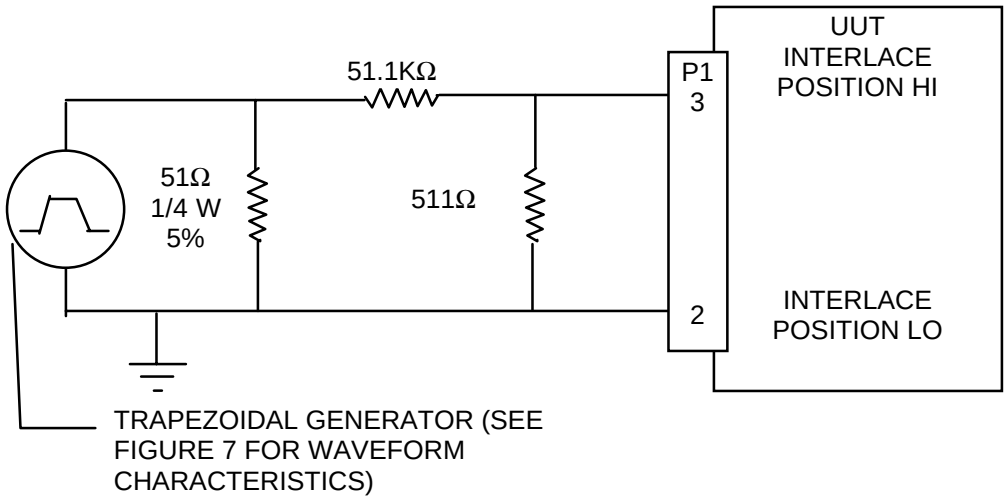


FIGURE 6. Test Interface

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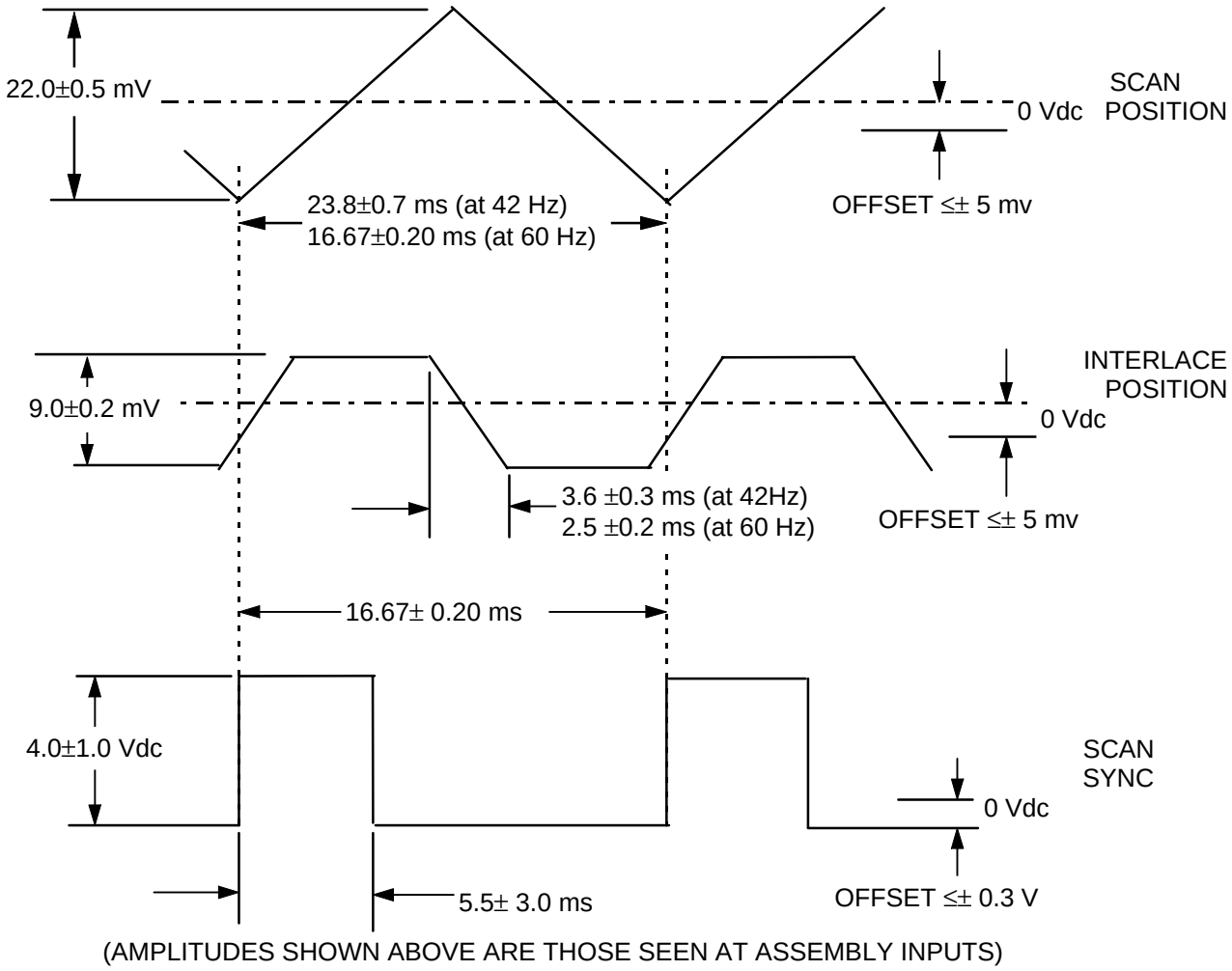


FIGURE 7. Input Signal Waveforms

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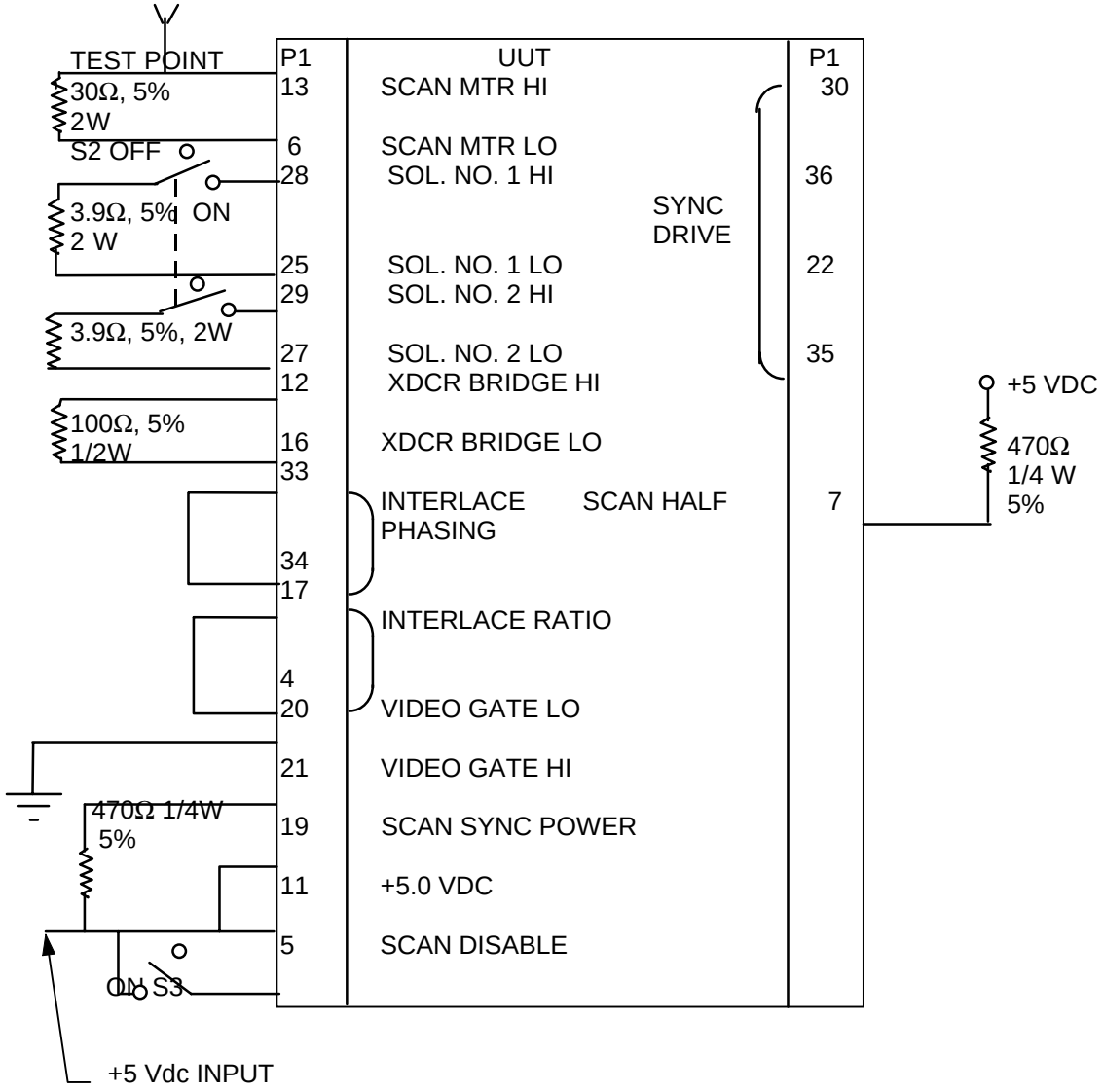
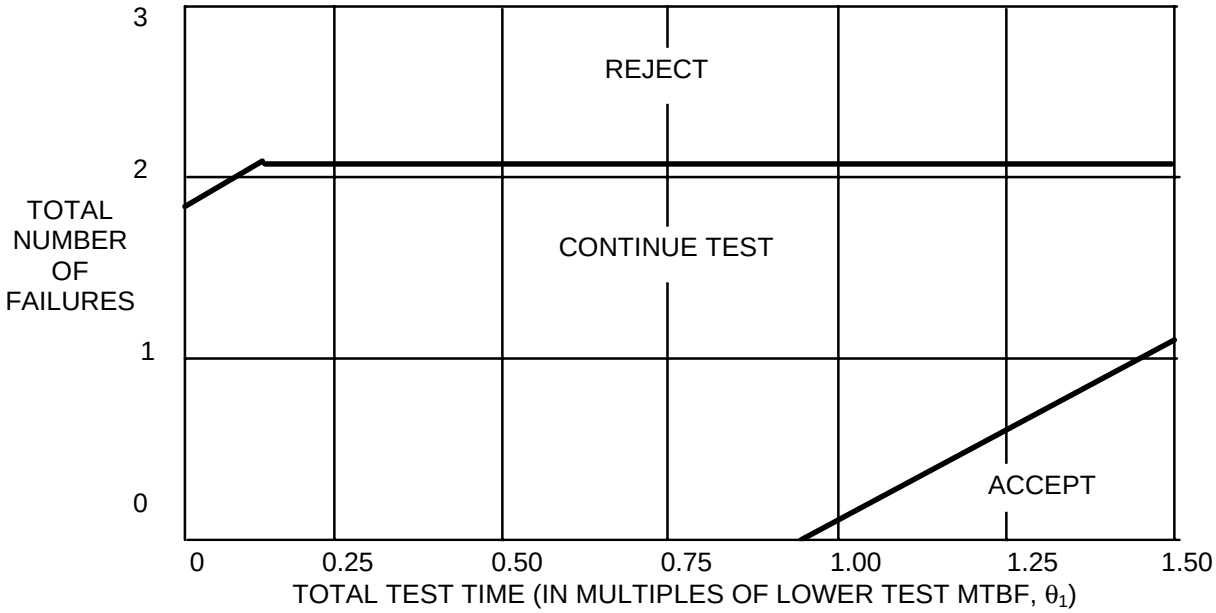


FIGURE 8. Test Interface

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TOTAL TEST TIME*

NUMBER OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF EQUIPMENT “ON TIME” AND IS EXPRESSED IN MULTIPLES OF LOWER TEST MTBF θ_1

FIGURE 9. First Article Reliability Accept-Reject Criteria

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49177B	**2. DOCUMENT DATE (YYMMDD)** 970519

3. DOCUMENT TITLE SCAN-INTERLACE, PL-1408/UA

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE	b. TELEPHONE *Include Area Code)* (1) Commercial (732) 532-2735 (2) AUTOVON 992-2735
c. ADDRESS *(Include Zip Code)* US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, ATTN: AMSEL-LC-LEO-E-EP FORT MONMOUTH, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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